

Hidden vegetables: an effective strategy to reduce energy intake and increase vegetable intake in adults

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Background: The overconsumption of energy-dense foods leads to excessive energy intakes. The substitution of low-energy-dense vegetables for foods higher in energy density can help decrease energy intakes but may be difficult to implement if individuals dislike the taste of vegetables. Objective: We investigated whether incorporating puréed vegetables to decrease the energy density of entrées at multiple meals reduced daily energy intakes and increased daily vegetable intakes. Design: In this crossover study, 20 men and 21 women ate ad libitum breakfast, lunch, and dinner in the laboratory once a week for 3 wk. Across conditions, entrées at meals varied in energy density from standard versions (100% condition) to reduced versions (85% and 75% conditions) by the covert incorporation of 3 or 4.5 times the amount of puréed vegetables. Entrées were accompanied by unmanipulated side dishes. Participants rated their hunger and fullness before and after meals. Results: Subjects consumed a consistent weight of foods across conditions of energy density; thus, the daily energy intake significantly decreased by 202 ± 60 kcal in the 85% condition ($P < 0.001$) and by 357 ± 47 kcal in the 75% condition ($P < 0.0001$). Daily vegetable consumption significantly increased from 270 ± 17 g of vegetables in the 100% condition to 487 ± 25 g of vegetables in the 75% condition ($P < 0.0001$). Despite the decreased energy intake, ratings of hunger and fullness did not significantly differ across conditions. Entrées were rated as similar in palatability across conditions. Conclusions: Large amounts of puréed vegetables can be incorporated into various foods to decrease the energy density. This strategy can lead to substantial reductions in energy intakes and increases in vegetable intakes. This trial was registered at clinicaltrials.gov as NCT01165086.